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ARTIFICIAL INTELLIGENCE

Course Portfolio

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Abstract

This portfolio documents my academic and technical learning experience throughout the Artificial Intelligence course during the Spring 2026 semester as part of my study abroad program. The portfolio includes programming assignments, technical reports, quizzes, essays, discussions, and personal reflections that demonstrate my evolving understanding of artificial intelligence concepts and methodologies. Topics explored throughout the semester include intelligent agents, search algorithms, constraint satisfaction problems, logical reasoning, machine learning concepts, Large Language Models, and hybrid AI systems. Through both theoretical analysis and practical implementation, this course strengthened my understanding of how AI systems operate, their limitations, and the importance of combining modern AI approaches with structured reasoning and ethical responsibility.

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1. Introduction

Artificial Intelligence has become one of the most influential technological fields in modern society. Throughout this course, I studied the principles behind intelligent computational agents, reasoning systems, search algorithms, and modern AI technologies. The course combined theoretical foundations with practical programming assignments and ethical discussions, allowing me to develop both technical and conceptual understanding of Artificial Intelligence.

This portfolio serves as a comprehensive representation of my learning experience during the semester. It includes technical reports, programming assignments, quizzes, essays, reflections, and collaborative work completed throughout the course. The portfolio also documents how my understanding of AI evolved, particularly regarding Large Language Models, hybrid systems, reasoning limitations, and the relationship between classical and modern AI approaches.

2. Course Overview

This portfolio documents my academic and technical learning experience throughout the Artificial Intelligence course (ICOM5015) during the Spring 2026 semester at the University of Puerto Rico Mayagüez as part of my study abroad program. The purpose of this portfolio is to present the assignments, technical reports, programming projects, quizzes, essays, discussions, and reflections completed during the course while demonstrating the development of my understanding of Artificial Intelligence concepts and applications.

The course provided an introduction to Artificial Intelligence from the perspective of designing and analyzing intelligent computational agents that can operate under limited information and computational resources. Throughout the semester, topics such as intelligent agents, search algorithms, constraint satisfaction problems, logical reasoning, knowledge representation, adversarial search, probabilistic reasoning, and introductory concepts related to generative AI and Large Language Models were studied both theoretically and practically.

In addition to programming assignments and technical implementations, the course emphasized critical thinking regarding the ethical, social, and philosophical implications of Artificial Intelligence. Discussions about intelligence, reasoning, explainability, hybrid AI systems, and the limitations of modern AI technologies contributed significantly to my understanding of how AI systems operate, where they succeed, and where they fail.

This portfolio reflects not only the technical skills developed during the semester, but also the evolution of my perspective on Artificial Intelligence as a field that combines computation, reasoning, ethics, and problem solving.

3. Course Outcomes

Recognize and apply AI techniques

Throughout the semester I worked with intelligent agents, search algorithms, logical reasoning systems, and constraint satisfaction problems. Programming assignments such as the Zebra CSP and Monte Carlo Briscas implementation helped reinforce these concepts.

Analyze and compare AI methods

The course emphasized comparing multiple AI approaches depending on the characteristics of the problem. Through discussions and assignments, I learned how different AI paradigms provide advantages and limitations depending on uncertainty, complexity, and computational constraints.

Design and implement AI systems

Several assignments required implementing AI-based solutions using Python and algorithmic reasoning techniques. These projects strengthened my practical understanding of AI development and experimentation.

Review AI literature and ethical implications

Technical reports and ethical discussions helped me critically analyze AI technologies beyond implementation. This included studying AI's societal impact, labor automation, explainability, and ethical concerns surrounding modern generative AI systems.

3. Programming Assignments

Zebra Constraint Satisfaction Problem

This assignment focused on solving the Zebra CSP using logical constraints and structured reasoning. It strengthened my understanding of constraint propagation, search spaces, and formal AI problem-solving techniques.

Monte Carlo Briscas Agent

The Monte Carlo assignment explored adversarial reasoning and probabilistic simulation techniques applied to a card game environment. This project demonstrated how stochastic methods can be used for decision-making in uncertain environments.

Python vs NumPy Performance Study

This technical assignment compared pure Python matrix multiplication implementations against NumPy optimized operations. The project reinforced the importance of computational efficiency, vectorization, and scalable numerical processing in AI systems.

Search and Intelligent Agents Assignments

Assignments related to vacuum agents and search algorithms helped demonstrate the role of rationality, perception, and environmental interaction in AI systems.

4. Technical Reports

The technical reports completed during the course followed formal engineering report structures that included introductions, theoretical background, methodology, analysis, conclusions, references, and appendices. These reports emphasized not only technical implementation, but also analytical reasoning, teamwork coordination, and professional scientific communication.

One significant report analyzed the computational performance differences between direct Python implementations and NumPy optimized numerical libraries. Another report explored intelligent agents and rationality in simulated environments. Both projects helped improve my technical writing and engineering analysis skills.

5. Essays and Discussions

The course included multiple philosophical and technical discussions regarding Artificial Intelligence and its societal implications. Topics included definitions of intelligence, AI ethics, robotics, agricultural AI applications, and navigation systems.

One important realization throughout these discussions was understanding that intelligence is not simply the ability to generate convincing responses, but also involves reasoning, planning, adaptability, memory, and contextual understanding. These discussions significantly influenced my understanding of the limitations and strengths of current AI systems.

6. Exams (Midterm and Final)

The midterm and final examinations served as comprehensive assessments of both the theoretical and practical concepts covered throughout the semester. I obtained a score of 98% on both examinations, which reflected my understanding of the material and my ability to apply AI concepts to analytical scenarios.

The midterm focused heavily on intelligent agents, search problem formulation, PEAS descriptions, rationality, environment properties, search strategies, and the emerging role of Large Language Models within AI systems. One of the most valuable aspects of the midterm was the analytical section, which required designing an AI academic advising assistant. Rather than simply recalling definitions, I had to formulate the problem using concepts such as state representations, actions, goal tests, path costs, and task environment classification. The exam also required proposing a hybrid architecture that combined symbolic planning with a language model interface and explaining the safeguards necessary to prevent errors before recommendations were shown to users.

The final examination expanded on these ideas by integrating concepts from Chapters 1 through 7 together with the course lessons on LLMs. The exam connected intelligent agents, search algorithms, heuristic search, adversarial reasoning, constraint satisfaction problems, logical agents, and modern generative AI systems. Several questions required analyzing AI systems that combined tool use, reasoning, and language generation, reinforcing the idea that modern AI applications are often composed of multiple interacting components rather than a single model.

Overall, the examinations helped consolidate my understanding of Artificial Intelligence as a multidisciplinary field that combines search, reasoning, planning, optimization, knowledge representation, and language technologies. More importantly, they reinforced the idea that successful AI systems require not only powerful models, but also mechanisms for validation, explainability, and reliable decision making.

7. Quizzes and Learning Progress

The quizzes throughout the semester helped reinforce the theoretical concepts covered in class. Initially, topics related to heuristic search methods and constraint satisfaction problems were challenging. However, continuous practice through assignments and discussions improved my understanding of logical inference, adversarial search, and reasoning systems.

The progression throughout the course demonstrated the importance of combining theoretical study with practical experimentation in order to fully understand AI methodologies.

8. Teamwork and Collaboration

Most programming assignments and technical reports were completed collaboratively as part of Team H. Teamwork involved task distribution, debugging, report writing, experimentation, and presentation preparation. Through collaborative work, I improved communication skills, technical coordination, and the ability to explain AI concepts clearly during demonstrations and discussions.

The collaborative nature of the projects also demonstrated the multidisciplinary character of Artificial Intelligence development and research.

9. Current AI Literature Review

Throughout the semester, several technical reports, articles, and research materials were reviewed to support assignments and class discussions. These materials helped connect theoretical concepts presented in class with real-world applications of Artificial Intelligence.

One of the most significant technical investigations conducted during the semester was the "Python vs NumPy Performance Study." This report analyzed the computational performance differences between matrix multiplication implemented using pure Python loops and optimized NumPy libraries. The results demonstrated the importance of computational efficiency and scalable numerical processing, which are essential considerations in modern AI and machine learning systems. This study reinforced the idea that successful AI applications depend not only on intelligent algorithms but also on efficient implementation.

Another important research activity was the "AI and Society: Ethical Impact Analysis" report. This project explored the effects of Artificial Intelligence on employment, privacy, surveillance, misinformation, healthcare, and scientific research. Through the application of consequentialist, deontological, and virtue ethics perspectives, I learned that AI technologies must be evaluated not only according to their technical capabilities but also according to their societal consequences and ethical implications.

These literature reviews and technical reports expanded my understanding of Artificial Intelligence beyond programming and algorithm design. They demonstrated that successful AI systems require careful consideration of efficiency, reliability, explainability, ethics, and human impact.

10. Applications and Demonstrations

The programming assignments completed throughout the semester provided practical experience in designing, implementing, and analyzing Artificial Intelligence systems.

These projects allowed me to apply theoretical concepts to realistic computational problems while developing problem-solving and programming skills.

The Zebra Constraint Satisfaction Problem demonstrated how complex logical problems can be represented using variables, domains, and constraints. Through this assignment, I gained practical experience with constraint satisfaction techniques and structured reasoning approaches that are commonly used in scheduling, planning, and optimization problems.

The Monte Carlo Briscas Agent project introduced probabilistic reasoning and decision-making under uncertainty. By using simulation-based techniques to evaluate potential actions, the project illustrated how AI systems can operate effectively even when complete information is unavailable.

The intelligent agents assignments explored the concepts of rationality, perception, and environmental interaction. Through the implementation of vacuum-world agents and related simulations, I learned how agent behavior can be evaluated according to performance measures and environmental conditions.

Finally, the Python versus NumPy performance study demonstrated the importance of computational efficiency in AI systems. The project showed that algorithm implementation choices can significantly affect performance and scalability, particularly when working with large datasets and computationally intensive operations.

These applications and demonstrations helped bridge the gap between theoretical knowledge and practical implementation. They also highlighted the diverse range of techniques used in Artificial Intelligence, from logical reasoning and search methods to probabilistic simulation and computational optimization.

11. Self-Assessment

Throughout the course, I developed a stronger conceptual understanding of intelligent agents, reasoning systems, search algorithms, and modern AI applications. I also improved my technical communication and analytical skills through formal reports and presentations.

One of my strengths throughout the semester was connecting theoretical AI concepts with practical applications and ethical considerations. Areas where I would like to continue improving include probabilistic reasoning, advanced optimization techniques, and deeper mathematical foundations behind machine learning systems.

12. Final Reflection Addendum

At the beginning of the course, I viewed Artificial Intelligence primarily as systems capable of generating intelligent behavior or solving problems efficiently. However, after studying intelligent agents, search methods, reasoning systems, and especially Large Language Models, my understanding of AI became much more nuanced.

Large Language Models demonstrated impressive capabilities in generating human-like language and assisting with reasoning tasks. However, the course also revealed important limitations of these systems, including hallucinations, lack of true understanding, inconsistent reasoning, and difficulties with explainability and validation.

Studying classical AI methods such as search algorithms, rational agents, and rule-based systems showed me that structured reasoning approaches provide reliability, interpretability, and validation mechanisms that purely statistical models often lack. This helped me understand why hybrid AI systems are increasingly important.

Hybrid systems that combine Large Language Models with structured reasoning, memory, validation systems, and symbolic AI approaches appear to provide a more reliable direction for future AI development. This course ultimately changed my understanding of AI from simply “intelligent behavior” to a broader perspective involving reasoning, ethics, accountability, and system design.

13. Appendix

Github link: <https://github.com/clauusanchezz/ICOM5015-AI-Portfolio-ClaudiaSanchez>

AI Acknowledgment

I used ChatGPT as an AI-assisted learning and writing support tool for brainstorming ideas, improving wording, organizing reflections, and reviewing concepts related to artificial intelligence. The final interpretations, reflections, conclusions, and organization of this portfolio are my own. I reviewed and edited all AI-assisted content to ensure accuracy, originality, and consistency with the course concepts and academic integrity expectations.